

Program: Diploma in Civil and Rural Engineering	
Course Code: 3368	Course Title: Construction Technology Lab
Semester: 3	Credits: 1.5
Course Category:	
Periods per week: 3 (L:0 T:0 P:3)	Periods per semester: 45

Course Objectives:

- To impart knowledge to the students to identify the physical properties of cement, fine aggregate, and coarse aggregate by performing laboratory tests
- To enable students to prepare concrete of desired compressive strength and specifications by maintaining the quality of concrete
- To provide hands-on experience for the students with the construction materials by performing masonry construction, and painting activities.

Course Prerequisites:

Topic	Course code	Course name	Semester

Course Outcomes:

On completion of the course, the student will be able to:

CO _n	Description	Duration (Hours)	Cognitive level
CO1	Assess the different properties of Cement.	9	Applying
CO2	Determine the different properties of aggregates.	9	Applying
CO3	Design & describe the preparation of green concrete and testing of concrete and determine the properties of hardened concrete.	12	Applying
CO4	Perform setting out of a building and demonstrate brick masonry and Carry out the painting works on different surfaces	11	Applying
	Lab Tests	4	

CO-PO Mapping:

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	3			3			
CO2	3			3			
CO3	3			3			
CO4	3			3			

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Assess the different properties of Cement.		
M1.01	Determination of fineness and consistency of cement.	2	Applying
M1.02	Determination of setting time of cement	2	Applying
M1.03	Determination of specific gravity of cement	2	Applying
M1.04	Determination of compressive strength of cement.	3	Applying
CO2	Determine the different properties of aggregates.		
M2.01	Determination of fineness modulus of fine aggregate and coarse aggregate	3	Applying
M2.02	Determination of specific gravity of fine aggregate and coarse aggregate.	3	Applying
M2.03	Determine bulk density of fine aggregate and coarse aggregate.	3	Applying
	Lab test	2	
CO3	Design & describe the preparation of green concrete and testing of concrete and determine the properties of hardened concrete.		
M3.01	Determination of workability of concrete by slump cone test.	3	Applying
M3.02	Determination of workability of concrete by compaction factor apparatus	3	Applying

M3.03	Determination of compressive strength of concrete.	3	Applying
M3.04	Determination of split tensile strength of concrete.	3	Applying
CO4	Perform setting out of a building and demonstrate brick masonry and Carry out the painting works on different surfaces		
M4.01	Measure dimensions of 10 bricks and find average dimension, weight & density. Perform field tests - dropping, striking and scratching by nail and correlate the results obtained.	3	Applying
M4.02	Construct one and a half brick masonry in English bond and Flemish bond	3	Applying
M4.03	Perform white washing and Colour washing, painting with enamel and emulsion paints.	2	Applying
M4.04	Setting out a small building in the field. Open ended project	3	Applying
	Lab test	2	

Text / Reference:

T/R	Book Title/Author
T	M. L. Gambhir, Concrete Technology, Mcgraw Hill Education.
R1	Civil Engineering Construction Materials, S.K. Sharma, Khanna Publishing House, New Delhi
R2	Concrete Technology, Shetty, M.S, S. Chand and Co. Pvt. Ltd., Ram Nagar, Delhi
R3	Laboratory Manual in Concrete Technology, Sood, H., Kulkarni P. D., Mittal L. N., CBS Publishers, New Delhi.
R4	Engineering Materials, Rangwala, S.C, Charator publisher, Ahemdabad
R5	Concrete Technology, Neville, A. M , Pearson Education Pvt. Ltd., New Delhi.

Online resources:

Sl.No.	Website Link
1.	https://civiltoday.com/civil-engineering-materials/cement/111-properties-of-cement-physical-chemical-properties
2.	https://civilread.com/14-different-types-cement-must-know/

3.	https://theconstructor.org/tips/setting-out-a-building-plan-on-ground/5897/
4.	https://theconstructor.org/building/types-bonds-brick-masonry-flemish-english-wall/11616/